appreciable length of time in certain specific states only, which are characterized by specific electron orbits—in other words, by definite energy levels of their planetary electrons—and that when they are in these stationary states, they do not radiate. This theory explained the stability of atoms, but there was still a long way to go before the natural laws governing atomic structure could become known.

During the first decade of the existence of Bohr's theory, the chemical properties of the elements were finally explained on the ground of the quantum theory. A further important step was successfully taken by the Frenchman Louis de Broglie, who realized, in 1924, that this strange duality of character which at times gave light the aspect of a wave, at times that of a swarm of particles, was a property not only of light, but of matter, too. This discovery finally led to the formulation of wave or quantum mechanics, which in a sense brought the theory of the extranuclear structure of the atom to completion.

Let us now try to elucidate this so-called duality of matter by two photographs. To begin with, let us observe once again the tracks of alpha particles in Figure 3, where their particle character is most clearly evident. After a study of these tracks, one can no longer have the slightest doubt that very small particles have actually flown through space and been deflected at some point here.

On the other hand, we have the evidence of experiments which indicate, with the same degree of certainty, that alpha rays are not particles but waves propagated from the source of radiation. I will not demonstrate this by the alpha rays, but by the beta rays, the particle character of which has been proved by numerous experiments no less convincingly than that of alpha rays. If beta rays are allowed to penetrate thin layers of matter, they show exactly the same kind of interference phenomena as may be observed in the case of x-rays under the same conditions—and we are accustomed to regard x-rays as typical wave radiation. The interference phenomenon consists in the fact that a central ray which penetrates the layer of matter, is surrounded by interference rings, which can be interpreted only as a superimposition of deflected waves. For comparison, we show here two photographs, one (Figure 5) taken by x-rays, and another (Figure 6) taken by beta rays. The latter one is one of the first photographs of this kind ever taken. But for the difference in their sharpness these two photographs would be indistinguishable.

As we see, it is impossible to entertain any doubt whatsoever as to the remarkable dual character of the electron. On the one hand, electrons may be regarded quite legitimately as particles, and we can be absolutely certain that a snapshot of an atom will be a picture similar to that shown in Figure 1. But on the other hand, they appear as waves, too, and this wave nature can also be used to obtain a picture of the atom—although in this case, the picture will